INCH-POUND
MIL-T-24742(SH)
26 April 1990

MILITARY SPECIFICATION

TRANSDUCER, PRESSURE AND DIFFERENTIAL PRESSURE, MINIATURE (ELECTRICAL) (NAVAL SHIPBOARD USE)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for miniature, electrical pressure and differential pressure transducers for Naval ships, but does not include the readout or display.

1.2 Classification.

1.2.1 Designations. Designations shall be assigned as specified (see 6.2) and listed in the format below:

M24742-D -	F -	15C -	5 -	DC -	A
Type	Application	Range	Pressure	Input Power	Output
1.2.1.1	1.2.1.2	1.2.1.3	1.2.1.4	1.2.1.5	1.2.1.6

1.2.1.1 Type. The transducer type shall be designated by the single letter symbol as follows:

D - Pressure, differential
R - Pressure, (gauge, absolute and sealed)
V - Pressure, vacuum
C - Pressure, compound (gauge, vacuum)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 6685
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

1.2.1.2 Application. The media pressure to be measured shall be indicated as follows:

F - Freshwater, condensate, steam, nitrogen and other inert gases
S - Seawater
G - Flue gas and ammonia
O - Oxygen
H - Liquid hydrocarbons (for example: fuel oil, lube oil)

1.2.1.3 Range. The pressure range of the transducer shall be designated by two parts. The first part shall be the designator for the numerical value of the upper range except 30 for type V (see 3.3.3). The second part shall be the designator for the upper range unit of measure (see 3.3.3.1).

1.2.1.4 Pressure rating. The pressure rating shall be designated by the designator for its numerical value for type D transducers ("X" for type R, V and C transducers) and shall be limited to the following:

PRESSURE RATING (POUNDS PER SQUARE INCH) TRANSDUCER TYPE "D"	
~~Range~~ Rating	Designator
15	1
150	2
300	3
600	4
1500	5
3000	6
6000	7

1.2.1.5 Power supply. The transducer wiring shall be determined by the power supply as follows:

dc - Two-wire system used with a direct current (dc) supply
ac - Four-wire system used with an alternating current (ac) supply

1.2.1.6 Output. The electrical signal output of the transducer shall be designated by the following symbols (see 3.4.4):

A - Milliampere (mA)
V - Volt (V)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917	- Electric Power Equipment, Basic Requirements for, (Naval Shipboard Use).
MS3102	- Connector, Receptacle, Electric, Box Mounting.
MS3106	- Connector, Plug, Electric, Straight, Solder Contacts, AN Type.
MIL-C-5015	- Connector, Electrical, Circular Threaded, General Specification for.
MS16142	- Boss, Gasket and Seal Straight Thread Tube Fitting, Standard Dimensions for.
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.
MIL-L-19140	- Lumber and Plywood, Fire-Retardant Treated.
MIL-S-19500	- Semiconductor Devices, General Specification for.
MIL-P-24423	- Propulsion and Auxiliary Control Consoles and Associated Control and Instrumentation Equipment, Naval Shipboard Use, Basic Design Requirements.
MIL-M-38510	- Microcircuits, General Specification for.
MIL-C-52211	- Components and Assemblies for Industrial Gas Production, Storage and Transport Equipment, Packaging of.

STANDARDS

MILITARY

MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-278	- Welding and Casting Standard.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-701	- Lists of Standard Semiconductor Devices.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
NAVSHIPS 810-1385846 - Oxygen System, Submarine Replenishment.

(Application for copies of NAVSEA drawing should be addressed to: Commander, Portsmouth Naval Shipyard, Naval Engineering Drawing Support Activity, Code 202.2, Portsmouth, NH 03804-5000.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

INSTRUMENT SOCIETY OF AMERICA (ISA)
Recommended Practice 37.1 - Nomenclature and Specification Terminology for Aerospace Test Transducers with Electrical Output.

(Application for copies should be addressed to the Instrument Society of America, P.O. Box 12277, Research Triangle Park, NC 27709.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4), in accordance with 4.3.

3.2 Description. The equipment specified herein shall convert a fluid pressure or differential pressure to a continuous linear proportional analog electrical dc signal. The transducer shall consist of the following components:

(a) Detector to sense the fluid pressure or differential pressure.
(b) Input power regulator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

(c) Signal conditioner to convert the detector output to a continuous linear proportional analog electrical dc signal.

3.3 General requirements. Transducer equipment shall be in accordance with provisions as specified herein. The transducer components shall be packaged as a single unit.

3.3.1 Materials. Materials shall be in accordance with the prohibited and galvanic corrosion material requirements of MIL-E-917 and MIL-P-24423, and shall conform to requirement 4 of MIL-STD-454 for fungus inertness.

3.3.2 Sensing elements. The materials for the force summing devices and wetted parts shall be in accordance with table I.

TABLE I. Material.

Sensing element and wetted parts	Service fluid				
	Fresh water, condensate, steam, nitrogen and other inert gases	Seawater	Flue gas and ammonia	Oxygen	Liquid Hydro-carbons (e.g., lube oil, fuel oil)
CRES 304L 316 L and 321	X	---	X	---	X
CRES 15-5 PH 17-4 PH 17-7 PH	 X X	 --- ---	 --- ---	 --- ---	 X X
Monel and K-Monel	X	X	---	X	X
Inconel 600 and 750	X	X	---	---	X
Inconel 625 and 718	X	X	---	---	X
Hastelloy C276	X	X	X	---	X
Titanium CP and 6Al-4V	X	X	---	---	X
CuNi 70/30	X	X	---	X	X
Ni Span	X	---	---	---	X
Tantalum	X	X	---	---	X

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

3.3.3 Ranges. Transducer pressure ranges shall be in accordance with table II.

3.3.3.1 Units. The units shall be designated by the corresponding letter designator and are limited to the following:

Designator	Units
A	HG - Inches of mercury vacuum
B	PSIA - Pounds per square inch, absolute
C	PSID - Pounds per square inch, differential
D	PSIG - Pounds per square inch, gauge
E	PSIS - Pounds per square inch, sealed at 14.7 PSIA
F	WC - Inches of water column
G	WCD - Inches of water column, differential

TABLE II. Ranges.

Type D				Type R				Type C		Type V	
Differential pressure ranges (PSID)		Differential pressure water column ranges (WCD)		Pressure ranges (PSIG, PSIA or PSIS*)		Water column ranges (WC)		Compound ranges (HG-0-PSIG)		Vacuum range (HG)	
Range	Designator	Range	Designator	Range	Designator	Range	Designator	Range	Designator	Range	Designator
0-15	11	0-10	21	0-15	31	0-10	71	30-0-15	51	0-30	30
0-30	12	0-60	22	0-30	32	0-60	72	30-0-30	52		
0-60	13	0-150	23	0-60	33	0-150	73	30-0-100	53		
0-100	14	0-300	24	0-100	34	0-300	74	30-0-150	54		
0-200	15			0-200	35			30-0-300	55		
0-400	16			0-300	36			30-0-600	56		
0-600	17			0-600	37						
				0-1000	38						
				0-1500	39						
				0-3000	41						
				0-6000	42						
				0-10,000	43						

* For upper range values of 1000 pounds per square inch (lb/in^2) and above.

3.3.4 Output. The electrical signal output of the transducer shall be dc directly proportional to the pressure or differential pressure input. The output shall be a true current source of 4 to 20 milliamperes (mA) dc or true voltage source of 0 to 5 volts (V) dc.

3.3.4.1 Current output. When a 4 to 20 mA current output is specified, the requirements specified herein shall be met regardless of external load resistance variations over a range of 0 to 250 ohms. The 4 mA output shall correspond to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

lower pressure or differential pressure range value and the 20 mA output shall correspond to the upper pressure or differential pressure range value for the ranges specified in table II.

3.3.4.2 Voltage output. When a 0 to 5 volt output is specified, the requirements specified herein shall be met for external load resistances exceeding 100 thousand ohms. The 0 volt output shall correspond to the lower pressure or differential pressure range value and the 5 volt output shall correspond to the upper pressure or differential pressure range value for the ranges specified in table II.

3.4 Transducer general requirements.

3.4.1 Life. The transducer shall be constructed for a life of 40,000 hours of operation and shall meet the requirements specified herein when subjected to the following conditions encountered in Naval service:

(a) Inclination (see 4.7.6).
(b) Supply voltage and frequency variations (see 4.7.7 and 4.7.10).
(c) Common mode pressure at pressure rating (see 4.7.8), type D only.
(d) Temperature (see 4.7.11) transducer ambient temperature operating range: 0 to 65 degrees Celsius (°C).
(e) Humidity (see 4.7.12).
(f) Pressure and differential pressure variations in the fluids from over range specified (see 4.7.17).
(g) Vibration (see 4.7.19).
(h) High shock (see 4.7.20).

3.4.2 Adjustments. Non-exposed adjustments for zero and span shall be provided for calibration purposes.

3.4.3 Enclosure. The transducer shall be splash proof as specified in MIL-STD-108.

3.4.4 Transducer mounting. The transducer shall have a mounting plate as shown on figure 1. If required in a specific application and with prior approval of NAVSEA, the transducer may be mounted by its pressure piping connection. For type D, the high pressure port shall be used. If the transducer is mounted by its pressure port, the mounting plate may be removed (see 6.2).

3.4.5 External configuration. The transducer shall have an external configuration conforming to figure 1.

3.4.6 Electrical connector. An electrical interface connector receptacle and mating plug, in accordance with MIL-C-5015 shall be provided with each transducer. The electrical receptacle may be designed to include built-in radio frequency suppression elements.

3.4.6.1 DC power input - A output. The receptacle shall be an MS3102R14S-5P in accordance with MS3102. Receptacle pin "A" shall be positive 28 Vdc power input, pin "B" shall be negative 28 Vdc power input, and pin "C" shall be case ground. Receptacle pins "A" and "B" shall also serve as the 4-20 mA signal output. The mating plug shall be an MS3106F14S-5S in accordance with MS3106.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

3.4.6.2 DC power input - V output. The receptacle shall be an MS3102R14S-5P in accordance with MS3102. Receptacle pin "A" shall be positive 28 Vdc power input, pin "B" shall be negative 28 Vdc power input, pin "C" shall be case ground, pin "D" shall be positive 0-5 Vdc signal output, and pin "E" shall be negative 0-5 Vdc signal output. The mating plug shall be an MS3106F14S-5S in accordance with MS3106.

3.4.6.3 AC power input - A output. The receptacle shall be an MS3102R14S-5PX in accordance with MS3102. Receptacle pins "A" and "B" shall be 115 Vac power input, pin "C" shall be case ground, pin "D" shall be positive 4-20 mA dc signal output, and pin "E" shall be negative 4-20 mA dc signal output. The mating plug shall be an MS3106F14S-5SX in accordance with MS3106.

3.4.6.4 AC power input - V output. The receptacle shall be an MS3102R14S-5PX in accordance with MS3102. Receptacle pins "A" and "B" shall be 115 Vac power input, pin "C" shall be case ground, pin "D" shall be positive 0-5 Vdc signal output and pin "E" shall be negative 0-5 Vdc signal output. The mating plug shall be an MS3106F14S-5SX in accordance with MS3106.

3.4.7 Pressure connections. Transducer pressure sensing connections for all services shall be a 7/16-20 UNF-2B tube connection in accordance with MS16142. However, for use on submarine oxygen replenishment systems shown on Drawing 810-1385846, the transducer sensing connections shall be a nickel-copper pipe nipple 1/4-inch nominal pipe size (nps) with 0.120-inch minimum wall thickness, 6 inches long, welded to the socket in accordance with MIL-STD-278 (see 6.2). For type D transducers the high and low pressure connections shall be clearly marked on the transducer body adjacent to the connections. The high pressure connection shall be on the end and the low pressure connection shall be on the side (see figure 1).

3.4.8 Input power requirements. Nominal steady-state power supply requirements for ac shall be 115 ± 8 volts, 60 ± 2 Hertz (Hz), single-phase. Nominal steady-state power supply requirements for dc shall be 28 ± 4.5 volts. The transducer shall operate with power supply variations as specified in 4.7.7. and 4.7.10.

3.4.9 Transducer cleaning. Prior to packaging for delivery, the contractor shall ensure that pressure transducers shall be free of all loose scale, rust, grit, filings and other foreign substances and free of mercury, oil, grease or other organic materials. The surfaces to be cleaned shall be the pressure connection and all parts contacted by the measurement fluid, except for oxygen where "clean" shall include all surfaces of the transducer. In addition, the following shall apply:

(a) Transducer for application O oxygen service (see 1.2.1.2) shall be clean gas calibrated, cleaned and packaged in accordance with MIL-C-52211.
(b) Transducers for all other applications shall be fresh water or clean gas calibrated, cleaned, and pressure connections capped.

3.4.9.1 Special marking. Transducers for use with applications F and O (see 1.2.1.2) shall have "USE NO OIL FOR CALIBRATION" prominently marked on the body.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

3.4.10 Welding and bolting. For application 0, all joints in the pressure system shall be welded or bolted.

3.4.11 Lubrication. The transducer shall operate without lubrication of moving parts after assembly. The legend "DO NOT LUBRICATE" shall be prominently marked on the body.

3.4.12 Damping. The use of oil for damping, as a fill media, and so forth, in transducers shall be cited on the equipment drawing.

3.4.13 Weight. The weight of a transducer shall not exceed 18 ounces.

3.4.14 Label plates. A metallic label plate with engraved or stamped markings shall be permanently affixed to the transducer, at a minimum, it shall contain:

(a) Pressure (or differential pressure) transducer.
(b) Manufacturer's name.
(c) Manufacturer's part number.
(d) Serial number.
(e) Contract number.
(f) Date of manufacture.
(g) Government designation (see 1.2.1).

3.5 Detail requirements.

3.5.1 Fuses. The transducer shall not be fused.

3.5.2 Semiconductors. Silicon rectifiers, diodes, transistors and other semiconductor devices shall be in accordance with MIL-S-19500 as applicable and shall be selected as specified in MIL-STD-701. Devices other than those listed in MIL-STD-701 shall be used only where specifically approved in writing by NAVSEA on the basis of justification submitted in accordance with the requirements of MIL-M-38510.

3.5.3 Electron tubes. Electron tubes shall not be used.

3.5.4 Resistors. Composition type fixed or variable resistors shall not be used.

3.6 Transducer performance. Unless otherwise specified, performance tolerances are specified in percent of transducer output span.

3.6.1 Static error band. The transducer static error band shall not exceed plus or minus 1.0 percent including thermal effect over the temperature range of 0 to 65°C (see 4.7.1).

3.6.2 Repeatability. Repeatability of transducer output shall be within 0.5 percent (see 4.7.1).

3.6.3 Sensitivity factor. The values of the ratio of electrical output percentage change to pressure input percentage change shall not be less than 0.75 or more than 1.25 (see 4.7.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

3.6.4 Ripple. Transducer root mean square (rms) output ripple shall not exceed 0.15 percent of full scale output (see 4.7.4).

3.6.5 Warm-up time. The transducer output shall attain a value within plus or minus 1.0 percent of the steady-state output with no overshoot in excess of 1.0 percent. Output shall reach this band in 1 minute or less after the transducer is energized and shall remain in this band (see 4.7.5).

3.6.6 Inclination. Maximum deviation of the transducer output resulting from inclination shall not exceed 3 percent (see 4.7.6).

3.6.7 Supply voltage and frequency or supply voltage (for dc) (steady-state). Maximum difference between outputs at any voltage and frequency or voltage (for dc) condition and the normal (115 volts, 60 Hz or 28 volts dc (Vdc)) at the same input and test temperature (differential pressure shall be included for type D) shall not exceed 1 percent (see 4.7.7).

3.6.8 Common mode pressure. During the common mode pressure test, for type D only, transducer performance shall be within the range formed by extending the upper and lower static error band limits specified in 3.6.1 by a percentage equal to

$$(1/10)\left[\frac{\text{(System pressure rating)}}{\text{Differential pressure range}}\right]^{1/3}$$

(see 4.7.8).

3.6.9 Response (see 4.7.9). Transducer output shall conform to the following criteria, where all percentages are of transducer span:

(a) Transducer output shall be within plus or minus 2 percent of maximum ramp pressure within 0.01 second of the time that pressure is attained.
(b) Transducer output shall exhibit no overshoot of maximum ramp pressure in excess of 2.0 percent.
(c) Transducer output shall indicate the actual pressure to within plus or minus 1 percent in 0.175 second or less after attainment of maximum ramp pressure, and shall remain within this error band for the duration of applied steady-state pressure.

3.6.10 Supply voltage and frequency (transient) (see 4.7.10).

3.6.10.1 Voltage. During the voltage transient test, the transducer output shall remain within plus or minus 2 percent of the pre-transient output.

3.6.10.2 Frequency. During the frequency transient test, the transducer output shall remain within plus or minus 0.5 percent of the steady-state output.

3.6.11 Temperature. During the temperature test, transducer performance shall be within the static error band specified 3.6.1 (see 4.7.11).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

3.6.12 Humidity. Maximum deviation of the second cycle reference measurement from the post-conditioning calibration shall not exceed 1.0 percent. After temperature cycling, performance shall conform to 3.6.1 and shall show no evidence of damage (see 4.7.12).

3.6.13 Overpressure. Calibration conducted after overpressure test shall have no values in excess of 1 percent deviation from the pre-overpressure test calibration. After calibration, the transducer shall adjust to the accuracy requirements of 3.6.1 (see 4.7.13).

3.6.14 Line pressure (type D only). After the line pressure test, transducer performance shall be within the static error band specified in 3.6.1 (see 4.7.14).

3.6.15 Salt spray. No appreciable corrosion or other damage, either mechanical or electrical, shall be evident after the salt spray test. After completion of the test, performance shall conform to 3.6.1 (see 4.7.16).

3.6.16 Pressure whip. Calibration conducted after completion of pressure whip test shall have no values in excess of 1 percent deviation from pretest reference measurement (see 4.7.17).

3.6.17 Insulation resistance. The insulation resistance of the transducer shall be not less than 10 megohms (see 4.7.18).

3.6.18 Vibration. Monitored transducer output during all phases of vibration test shall show no variation from steady-state output in excess of 2 percent. There shall be no visible evidence of damage to the transducer as a result of the vibration test (see 4.7.19).

3.6.19 Shock. After the shock test, but before any adjustment is made, the transducer output shall show no deviation greater than 3 percent. A post-test calibration, using adjustments provided, shall conform to performance requirements of 3.6.1. There shall be no visual evidence of damage to the transducer as a result of the shock test (see 4.7.20).

3.6.20 Burst pressure. The transducer shall withstand the burst pressure specified in 4.7.21 without showing any evidence of leakage.

3.6.21 Output. The output shall conform to 3.3.4 and the transducer performance shall be within the static error band specified in 3.6.1 (see 4.7.22).

3.6.22 Enclosure. There shall be no evidence of water leakage into the transducer (see 4.7.15).

3.6.23 Electromagnetic interference emission and susceptibility (EMI). The transducers shall meet the following requirements of MIL-STD-461: CE01, CE03, CS01, CS02, CS06, CS09, RE01, RS01, RS02, RS03, except as modified below (see 4.7.23):

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

(a) Requirement CS02 shall be met as specified except as follows:

(1) At frequencies between 2 and 30 megahertz, the full output of the generator (source impedence of 50 ohms, and an output capability of 12.25 volts root mean square, or 3 watts, when connected to a 50-ohm load) shall be applied to all power and signal leads of the test sample.
(2) The "calibrating resistor", or 50-ohm load, shown in the CS02 test block diagram of MIL-STD-462, shall not be connected during the test.
(3) The test sample output shall not vary by more than 0.5 percent of transducer span.
(4) The test signal shall be applied to all power and signal leads of the test sample.

(b) Requirement RS03 shall be met as specified except as follows:

(1) Between the frequencies of 14 kilohertz and 18 gigahertz, the electric field strength test level shall be 10 volts per meter (V/m).

(c) Requirement CS06 shall be met as specified except as follows:

(1) The test signal shall be applied only to ac power leads of the test sample.

(d) Requirements CE01 an CE03 shall be met as specified except as follows:

(1) The test signal shall be applied only to ac power leads of the test sample.

3.7 Workmanship. Workmanship of the transducers shall conform to requirement 9 of MIL-STD-454.

3.8 Drawings. (See 6.3 and appendix).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be performed prior to production. First article inspection shall be performed on samples which have been produced with equipment and procedures normally used in production. First article inspection shall consist of the examination and tests specified in table III.

TABLE III. First article examination and tests.

Examination and tests	Requirement	Examination and test
General examination	3.3	4.5
Output	3.6.21	4.7.22
Warm-up time	3.6.5	4.7.5
Enclosure	3.6.22	4.7.15
Salt spray	3.6.15	4.7.16
Static error band and repeatability	3.6.1 & 3.6.2	4.7.1
Sensitivity factor	3.6.3	4.7.3
Ripple	3.6.4	4.7.4
Inclination	3.6.6	4.7.6
Supply voltage and frequency (steady-state)	3.6.7	4.7.7
Common mode pressure (Type D only)	3.6.8	4.7.8
Response	3.6.9	4.7.9
Supply voltage and frequency (transient)	3.6.10	4.7.10
Temperature	3.6.11	4.7.11
Humidity	3.6.12	4.7.12
Overpressure	3.6.13	4.7.13
Line pressure (Type D only)	3.6.14	4.7.14
Pressure whip	3.6.16	4.7.17
Insulation resistance	3.6.17	4.7.18
Vibration	3.6.18	4.7.19
Shock	3.6.19	4.7.20
Burst pressure	3.6.20	4.7.21
Electromagnetic interference emission and susceptibility	3.6.23	4.7.23

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

4.3.1 Sample size. The number of samples to be subjected to first article inspection specified in table III shall depend on the transducer design. If each range is covered by a separate an distinct design, a sample for each range will require testing. In instances where a singular design series may cover several ranges and types, only three samples need be tested provided the electrical and mechanical similarities are approved by NAVSEA. In no case, however, shall less than three units, one unit each representing low (99 lb/in^2 and below), medium (100 to 999 lb/in^2) and high (1000 lb/in^2 and above) ranges, be tested, regardless of design similarity.

4.3.1.1 Order of inspection. With the exception of the electromagnetic interference emission and susceptibility test which may be conducted on separate transducers, the sample transducers shall be subjected to the inspections specified in table III in the order listed. Any deviation in the test order shall first be approved by NAVSEA.

4.4 Quality conformance inspection. Each pressure transducer in each lot offered for delivery shall be subjected to the inspection listed in table IV and shall be conducted in the order listed. Failure of any pressure transducer to meet the requirements of this specification shall be cause for rejection. An inspection lot shall consist of all pressure transducers of the same classification (see 1.2) produced under the same conditions and offered for delivery at the same time (see 6.3).

TABLE IV. Quality conformance inspection.

Inspection	Requirement	Examination and test
General examination	3.3	4.5
Output	3.6.21	4.7.22
Static error band and repeatability	3.6.1 and 3.6.2	4.7.1
Sensitivity factor	3.6.3	4.7.3
Overpressure	3.6.13	4.7.13
Line pressure (Type D only)	3.6.14	4.7.14
Insulation resistance	3.6.17	4.7.18

4.5 General examination. Each transducer shall be given a thorough examination to determine conformance to the requirements of this specification with respect to material, finish, workmanship, construction, assembly, dimensions, weight and marking of identification. Examination shall be limited to the examinations that may be performed without disassembling the units. Examination shall also include a check of all adjustments.

4.6 Test conditions. Except where the following factors are the variables, the tests specified in 4.7 shall be conducted with the equipment operating under the following conditions:

(a) Ambient temperature shall be 25 ± 2°C.
(b) Relative humidity shall be ambient.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

(c) Supply voltage shall be 115 volts (nominal) for input power designation ac or 28 volts (nominal) for input power designation dc.
(d) Supply frequency shall be 60 Hz (nominal) for input power designation ac or direct current for input power designation dc.
(e) Except where the external load is a test requirement, any value of resistance may be employed as follows:

For output designation A - between 0 and 250 ohms
For output designation V - above 100K ohms

4.7 Tests. Except for the warm-up time test (see 4.7.5), the transducer and all associated test equipment shall be energized for a period of time sufficient to ensure complete warm-up.

4.7.1 Static error band and repeatability. The transducer shall first be flexed over its full pressure range by slowly increasing and decreasing the applied pressure for six continuous cycles. The calibration measurement shall be made at a minimum of five equally spaced intervals over the entire range (both upscale and downscale). Precaution shall be taken to avoid overshoot. This calibration procedure shall be applied three successive times to determine repeatability. Static error band of all calibration shall meet the requirements of 3.6.1. Repeatability shall meet the requirements of 3.6.2.

4.7.2 Reference measurement. The following reference measurement shall be conducted when specified in the individual test:

(a) A one trial calibration with at least five equally spaced intervals over the entire transducer range both upscale and downscale.

4.7.3 Sensitivity factor. The sensitivity factor shall be determined as follows: Provide a pressure (differential pressure for type D) to the transducer to a level of 80 ± 5 percent of span. Record the input pressure (differential pressure) and corresponding electrical output. Increase the pressure (differential pressure) by an amount not exceeding 1 percent of span. Record both the new pressure (differential pressure) and corresponding new electrical output. Calculate the change in both applied pressure (differential pressure) and electrical output as a percentage of transducer span. Determine the ratio of electrical output percentage change to applied pressure (differential pressure) percent change. Repeat this procedure for a pressure (differential pressure) decrease not exceeding 1 percent of span. Performance shall conform to the requirements of 3.6.3.

4.7.4 Ripple. Rms ripple shall be determined at an input pressure (differential pressure for type D) of 80 ± 5 percent of transducer span. Performance shall conform to the requirements of 3.6.4.

4.7.5 Determination of warm-up time. Test shall be conducted to determine the elapsed time between the application of line power to the transducer and the point at which the transducer output reaches the conditions specified in 3.6.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

4.7.5.1 Test conditions. The transducer shall be placed in an ambient temperature of 25 ± 2°C for not less than 2 hours de-energize. Recording equipment and other auxiliary equipment shall be energized to ensure complete warm-up. An input pressure (differential pressure for type D) of 80 ± 5 percent of the transducer span shall be applied to the transducer and maintained constant during this test. Performance shall conform to 3.6.5.

4.7.6 Inclination. Prior to any inclination, a reference measurement in accordance with 4.7.2 shall be made with the transducer in each of the following orientations:

(a) Mounting surface horizontal.
(b) Mounting surface vertical with pressure connections (high pressure connection for type D) vertical downward.

4.7.6.1 From each of the mounting orientations of 4.7.6, the transducer shall be inclined for at least 1 minute in each of the following positions and a reference measurement in accordance with 4.7.2 shall be made. Performance shall conform to 3.6.6.

(a) 45 degrees from the vertical forward.
(b) 45 degrees from the vertical backward.
(c) 45 degrees from the vertical to the left.
(d) 45 degrees from the vertical to the right.

4.7.7 Supply voltage and frequency or supply voltage (for dc) (steady-state). The transducer shall be operated at the configurations of normal, maximum and minimum steady-state voltages. The ambient temperature shall also vary, with the transducer operated for at least 15 minutes at each test temperature before the first reference measurement. The transducer shall be allowed at least 15 minutes to stabilize at each configuration. After that a reference measurement shall be taken (see 4.7.2). Reference measurements shall be performed at ambient temperatures of 0 ± 2°C, 25 ± 2°C and 65 ± 2°C. Test temperatures shall be accomplished by varying temperature in steps of 10°C each (30 minutes for each step) until the desired ambient temperature is reached. Performance shall conform to 3.6.7.

4.7.8 Common mode pressure test (type D only). The transducer's rated pressure shall be applied simultaneously to both pressure ports. The pressure at the low pressure port shall then be decreased in pressure increments specified in 4.7.2 to the specified transducer range and then increased in similar increments to the transducer rated pressure. Performance shall conform to 3.6.8.

4.7.9 Response. A pressure (differential pressure for type D) ramp consisting of a pressure (differential pressure for type D) rise of at least 40 percent of transducer span occurring at a rate of not less than 400 percent per second shall be applied to the transducer. The maximum ramp pressure shall be maintained for at least 0.50 second and shall not vary by more than plus or minus 2 percent of the transducer span. Performance shall conform to 3.6.9.

4.7.10 Supply voltage and frequency or supply voltage (for dc) transient. Tests specified in 4.7.10.1 and 4.7.10.2 shall be conducted with a pressure (differential pressure for type D) input signal equal to 80 ± 5 percent of the transducer span. Performance shall conform to the requirements of 3.6.10.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

4.7.10.1 Transient voltage.

4.7.10.1.1 Upper limit of steady-state voltage. With the transducer operating at the upper limit of steady-state ac voltage, the ac powered transducer shall have a transient voltage of plus 15 volts recovering to the steady-state band in 2 seconds superimposed. With the transducer operating at the upper limit of steady-state dc voltage, the dc powered transducer shall have a transient voltage of plus 2.0 volts recovering to the steady-state band in 2 seconds superimposed.

4.7.10.1.2 Lower limit of steady-state voltage. With the transducer operating at the lower limit of steady-state ac voltage, the ac powered transducer shall have a transient voltage of minus 15 volts recovering to the steady-state band in 2 seconds superimposed. With the transducer operating at the lower limit of steady-state dc voltage, the dc powered transducer shall have a transient voltage of minus 2.0 volts recovering to the steady-state band in 2 seconds superimposed.

4.7.10.2 Transient frequency (for ac powered transducers).

4.7.10.2.1 Upper limit of steady-state frequency. With the transducer operating at the upper limit of steady-state frequency, a transient frequency of plus 1.5 Hz recovering to the steady-state band in 2 seconds shall be superimposed.

4.7.10.2.2 Lower limit of steady-state frequency. With the transducer operating at the lower limit of steady-state frequency, a transient frequency of minus 1.5 Hz recovering to the steady-state band in 2 seconds shall be superimposed.

4.7.11 Temperature. The transducer shall operate normally (without alignment or adjustment) throughout the following temperature cycle. Tolerances in operating characteristics shall be as specified herein.

(a) Hold test temperature at 0 ± 2°C for at least 24 hours.
(b) Increase test temperature in steps of 10 degrees each, at 30 minutes for each step, until plus 65 ± 2°C is reached and hold at that temperature for at least 4 hours.
(c) Reduce test temperature in steps of 10 degrees each, at 30 minutes for each step, until plus 25 ± 2°C is reached and hold at that temperature for at least 4 hours.

At each temperature plateau (0, 65 and 25°C), a reference measurement (see 4.7.2) shall be made. Performance shall conform to 3.6.11.

4.7.12 Humidity. Transducer shall be subjected to the following conditioning and test. Except for the periods of test indicated below, the equipment shall not be energized. Performance shall meet the requirements of 3.6.12.

4.7.12.1 Conditioning. In order to establish a reference condition for the measurement of operating parameters and a valid basis for comparison of the effects of the conditioning to follow, the complete equipment shall be dried at a temperature not less than 40°C or more than 50°C for at least 2 hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

4.7.12.2 Reference measurements. Following the conditioning specified in 4.7.12.1, a reference measurement in accordance with 4.7.2 shall be conducted at 25 ± 5°C and 50 ± 5 percent relative humidity to indicate satisfactory performance.

4.7.12.3 Temperature cycling. The transducer shall then be subjected to four 24-hour cycles of temperature variation consisting of not less than 18 hours at 65 ± 5°C and not less than 6 hours at 25 ± 5°C. The relative humidity shall be maintained above a minimum of 95 percent during the steady-state conditions. The transitions between temperatures shall be accomplished within the 6-hour period so that the time at the high temperature is not less than 18 hours. Each transition shall not exceed 1 hour if the transducer remains in the chamber, or 15 minutes if a two chamber method is employed. The relative humidity need not be controlled during the transition periods.

4.7.12.4 Measurement during cycling. During the second cycle, the measurement required in 4.7.12.2 shall be performed at 65 ± 5°C before the temperature is decreased to 25°C with the transducer remaining in the chamber. The transducer shall be energized for as brief a period as required to complete the measurements. A warm-up period may be permitted where previous tests indicate a definite period is required for the transducer to attain thermal stability.

4.7.12.4.1 Requirements of transducer. The transducer shall meet the requirements specified herein without alignment or adjustments, other than the accessible controls employed for operation of the equipment. No repairs shall be permitted before measurements are taken. If repairs shall be required, the test specified in 4.7.12.3 shall be repeated, after the necessary replacements have been made.

4.7.12.5 Measurements after temperature cycling. After the four complete cycles, the measurements required in 4.7.12.2 shall be performed at 25 ± 5°C with the transducer remaining in the chamber. The transducer shall meet the requirements of this specification with no readjustments.

4.7.12.5.1 Additional tests. Upon completion of the tests, and after remaining inoperative for not less than 12 hours nor more than 24 hours at a temperature of 25 ± 5°C and 50 ± 5 percent relative humidity, any additional tests or measurements considered necessary by the Government shall be made to determine conformance to the requirements specified herein. In addition, the transducer shall be examined to detect evidence of physical degradation such as corrosion of metal parts or distortion of plastic parts. When it is necessary to replace parts to obtain satisfactory transducer performance, the failed part or parts shall be analyzed to determine the cause of unsatisfactory operation. Unsatisfactory parts or materials shall be replaced by adequate substitutes.

4.7.13 Overpressure. Before the overpressure test, a reference measurement in accordance with 4.7.2 shall be made. The transducer shall successfully withstand pressure (differential pressure for type D) equal to 200 percent of its range with a maximum pressure of (differential pressure for type D) 12,000 lb/in^2 for a period of 1/2 hour. At the end of this period, transducers shall be immediately subjected to a pressure (differential pressure for type D) equal to 1 lb/in^2 or 10 percent of range, whichever is less, below atmospheric for an additional period of 1/2 hour. Within 10 minutes after release of this pressure

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

(differential pressure for type D), a reference measurement (see 4.7.2) shall be made for comparison. Performance shall conform to 3.6.13. For type D only, if the line pressure rating exceeds 200 percent of the differential pressure range, the overpressure test shall be omitted and the 0.5 percent deviation allowed in 3.6.13 shall be applied to the line pressure test (see 4.7.14).

4.7.14 Line pressure (for type D only). The transducer shall successfully withstand the pressure rating, when applied to the high pressure port with the low pressure port vented to atmosphere for a period of 10 minutes. The above shall be repeated with the pressure applied to the low pressure port of the transducer. After each test, a reference measurement in accordance with 4.7.2 shall be made. Performance shall conform to 3.6.14.

4.7.15 Enclosure. The enclosure shall be subjected to the splashproof test specified in MIL-STD-108. Performance shall conform to 3.6.22.

4.7.16 Salt spray. The salt spray test shall be conducted in accordance with method 101D of MIL-STD-202. Duration of the test shall be 96 hours. When any transducer component is physically attached to the detector, the integral component shall also be tested. If the transducer is not designed for seawater service, the inlet pressure connections shall be plugged. Performance shall conform to 3.6.15.

4.7.17 Pressure whip. Before performing the pressure whip test, a reference measurement shall be made (see 4.7.2). The test shall be conducted on a suitable system by applying a periodic pressure change of not more than 20 percent to not less than 80 percent of span for a total of 260,000 cycles. The rate of cycling shall be within the range of 0.25 to 2.0 Hz. The transducer shall be energized throughout the test. After completion of the pressure whip test, a reference measurement shall be made for comparison (see 4.7.2). Performance shall conform to 3.6.16.

4.7.18 Insulation resistance. The insulation resistance of the transducer shall be determined by applying 50 Vdc between electrical input and output circuits and between these circuits and ground. The temperature shall be 25 ± 5°C and the relative humidity shall be 50 ± 10 percent. The insulation resistance measurement shall be made immediately after a 2 minute period of uninterrupted test voltage application. However, if the indication of insulation resistance meets the specified limit (see 3.6.17) and is steady or increasing, the test may be terminated before the end of the 2-minute period.

4.7.19 Vibration. The transducer shall be tested in accordance with type I (environmental) vibration of MIL-STD-167-1 except that the upper frequency shall be 175 Hz; the amplitude of vibration shall be in accordance with table V; and for the variable frequency portion, the vibration level shall be maintained for 2 minutes at each integral value of frequency. If no resonances are observed, the 2 hour endurance test shall be conducted at 175 Hz. During the vibration test, a fluid pressure (differential pressure for type D) of 80 ± 5 percent of the transducer span shall be applied to the transducer. Transducer output during the test shall be monitored. Performance shall conform to 3.6.18.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

TABLE V. Amplitudes of vibration.

Frequency range (Hz)	Table amplitude (inches)	
	Exploratory	Variable frequency
5 to 20	0.010 ± 0.002	0.030 ± 0.006
21 to 50	0.004 ± 0.001	0.020 ± 0.004
51 to 100	0.0015 ± 0.0003	0.010 ± 0.002
101 to 175	0.0005 ± 0.0001	0.0015 ± 0.0003

4.7.20 Shock. The shock tests shall be conducted in accordance with grade A, class 1, type C using bulkhead mounting fixture 4-A of MIL-S-901. During the test, a fluid pressure (differential pressure for type D) of 80 ± 5 percent of the transducer span shall be applied to the transducer. Transducer output during the test shall be monitored. Before and after the shock test, reference measurements shall be made for comparison (see 4.7.2). Performance shall conform to 3.6.19.

4.7.21 Burst pressure. The transducer (pressure ports for type D) shall be subjected to a liquid pressure equal to 300 percent of the range with a maximum pressure of 15,000 lb/in^2 applied to the transducer (simultaneously to both sides for type D) for a period of 10 minutes. The transducer shall conform to the requirements of 3.6.20. No performance test shall be required after the burst pressure test. A reference measurement (see 4.7.2) shall be recorded for information purposes.

4.7.22 Output. A reference measurement shall be made in accordance with 4.7.2. Performance shall conform to the requirements of 3.6.21.

4.7.23 EMI tests. EMI tests shall be in accordance with the test methods specified in MIL-STD-462, with the modifications as specified in 3.6.23. Performance shall be as specified in 3.6.23.

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified herein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 Packaging requirements. Transducer equipment shall be individually preserved level A, C, or commercial, packed level A, B, C or commercial as specified (see 6.2) and marked including bar coding in accordance with MIL-E-17555 and shall include acquisition options therein as specified (see 6.2). Unless otherwise specified (see 6.2), method III shall apply for level A preservation. In addition, for Navy acquisitions, the following applies:

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

(a) Navy fire-retardant requirements.

(1) Lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container construction members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.
Level C - Type I - non-weather resistant.
Category 1 - general use.

(2) Fiberboard. Unless otherwise specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes including interior packaging forms shall meet the flamespread and the specific optic density requirements of PPP-F-320.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Transducers are intended to provide control input and monitoring of pressure and differential pressures for shipboard engineering plants.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Part classification required (see 1.2.1).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
(d) When the first article inspection is required (see 3.1).
(e) When mounting is by pressure connection (see 3.4.4).
(f) Type of pressure connection if other than specified herein (see 3.4.7).
(g) Quantity of transducer required.
(h) Level of preservation and level of packing required (see 5.1).
(i) Applicable acquisition requirements of MIL-E-17555 (see 5.1).
(j) When fire retardant material is not required (see 5.1(a)).
(k) Flamespread and specific optic density requirements, if other than as specified (see 5.1(a)(2)).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements,

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.8 and appendix	DI-DRPR-80651	Drawings, engineering	Level 3
4.4	DI-T-2072	Reports, test	---

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.3.1 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, Military specifications and standards which have been cleared and listed in DoD 5010.12-L (AMSDL) must be exhibited on a separate CDRL (DD Form 1423), included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract. When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.6 Sub-contracted material and parts. The packaging or preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

6.7 Definitions. Terminology consistent with Instrument Society of America (ISA) Recommended Practice 37.1 shall apply, except as modified by the definitions listed as follows:

(a) Static error band. Static error band is the maximum deviation from a straight line drawn through the coordinates of the lower range limit at specified transducer output, and the upper range limit as specified transducer output expressed in percent of transducer span.

(b) Common mode pressure. The common mode pressure is static line pressure applied simultaneously to both pressure sides of the transducer for the differential pressure (D) transducer only.

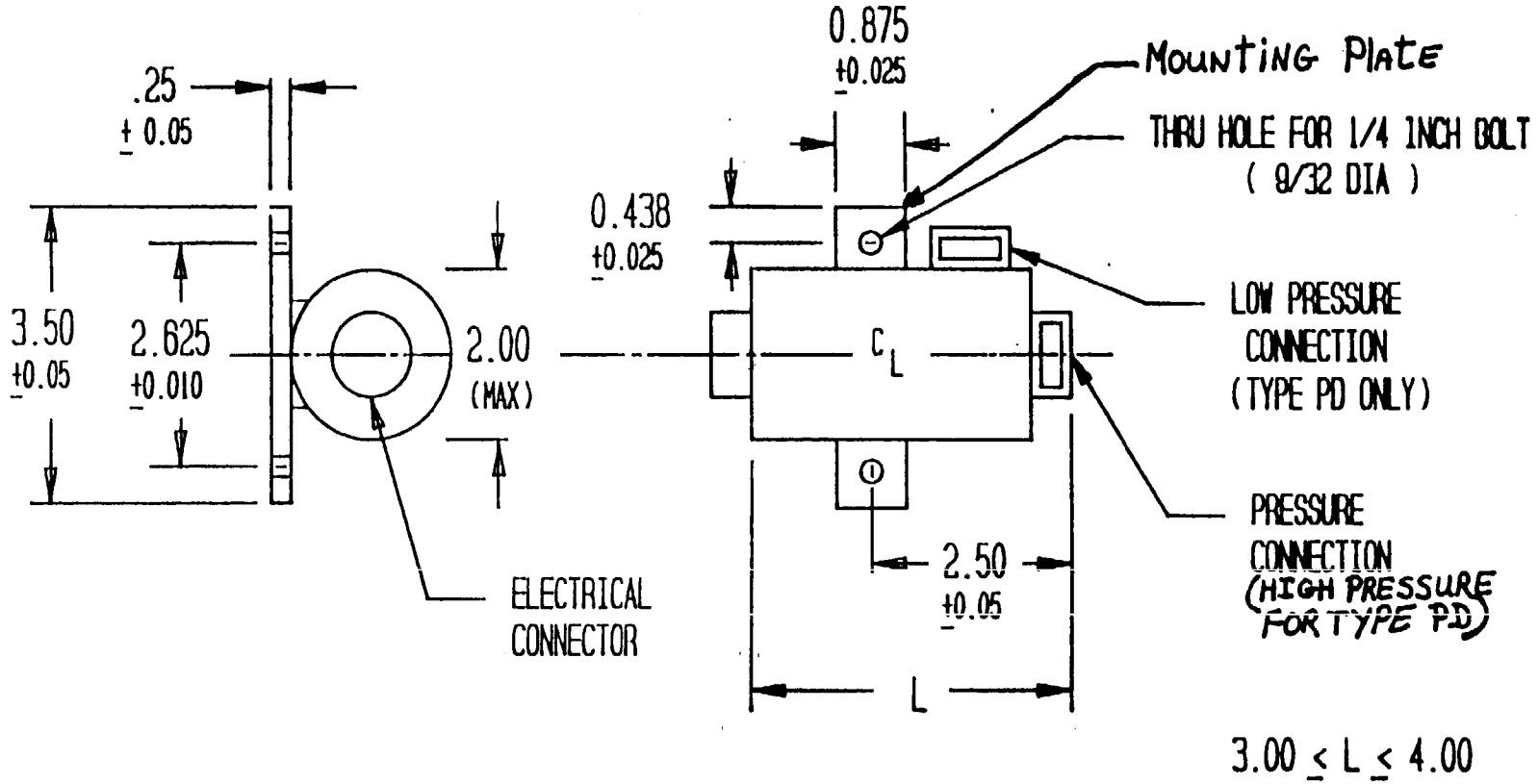
6.8 NATO coordination and standardization. The standard characteristics of the pressure transducer supplied for U.S. Navy ships have been coordinated with NATO standardization document, where applicable. In particular, the standard characteristics of the differential pressure transducer output conforms to corresponding transducer output specified in STANAG 4286.

6.9 Subject term (key word) listing.

Current
Fluid
Range

Preparing activity:
Navy - SH
(Project 6685-N823)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.



NOTES:

1. Transducer housing (body) cross section is shown as circular. Any alternate cross section not exceeding 2.00 inches in width and 2.00 inches in height is acceptable.
2. All dimensions are in inches.
3. Break all edges.

FIGURE 1. External configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

APPENDIX

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix contains the format and content preparation instructions for the development of and revision to a conformance verification drawing. It is not intended that each requirement contained herein should be applied to every type of instrumentation. Portions of this appendix are subject to deletion tailoring depending upon the material, construction and principle of operating requirements that are specified in the individual instrumentation specification or acquisition document. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423. The information contained herein is intended for compliance.

20. REFERENCED DOCUMENTS

20.1 Government documents. The following documents form a part of this appendix to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-I-45208 - Inspection System Requirements.

STANDARDS

MILITARY

MIL-STD-17-1 - Mechanical Symbols (other than aeronautical, aerospacecraft and spacecraft use) Part-1.

(Unless otherwise indicated, copies of federal and military specifications are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

20.1 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/Y14.1 - Drawing Sheet Size and Format.

ANSI/IEEE 200 - Standard Reference Designations for Electrical and Electronics Parts and Equipment.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

30. CONFORMANCE VERIFICATION DRAWING FOR INSTRUMENTATION

30.1 Purpose. The conformance verification drawing contains the information necessary to verify that the instrumentation meets the requirements specified in the applicable instrumentation specification and acquisition document.

30.2 Content and format. One conformance verification drawing shall be developed for each specific type of instrumentation. It shall include all ranges, sizes, connections and other variations. The conformance verification drawing shall include the following minimum information (except as specified in 10.1) and shall be developed to the following format:

30.3 Descriptive data.

(a) Instrument identification numbering system for instrumentation. This numbering system shall include, but may not be restricted to, the classification variables.
(b) Instrumentation identification number system for replaceable parts.
(c) Size, operating data, ranges, scale markings, and other data for proper selection.
(d) Test approval data, presenting the following information in tabular form.

(1) Specification classification (unique alphanumeric variables to designate instruments).
(2) Test report number and date.
(3) Facility where test was conducted.
(4) Authorized Government activity (NAVSEA) approval letter and date.

(e) A statement that instrumentation is in accordance with the requirements of the applicable instrumentation specification or acquisition document and to referenced specifications.
(f) Conformance verification drawing acceptance data, presenting the following information in tabular form:

(1) Authorized Government activity (NAVSEA) acceptance letter and date.
(2) Revision number.

30.4 Details of construction.

(a) Two or more representative assembly views, as required, to show clearly the details of the design, construction, and assembly of the instrumentation and to identify each part and its location. Identification of parts shall correspond to the list of materials. Assembly shall show how all mechanical parts are joined or attached.
(b) Sectional views or notes as necessary to show internal details.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

(c) Details such as entrance provisions, gaskets, fastening techniques, welding symbols, mounting requirements, and other details as applicable.
(d) Module enclosure requirements for separately mounted modules.
(e) Dimensions required to ensure interchangeability.
(f) Schematic of indicating system.
(g) Description of the nature and purpose of any adjustments.
(h) Welding procedures with acceptance data, including acceptance letter, report numbers and dates.
(i) Any special features.
(j) Complete weight.
(k) Location, size, and type of connections.
(l) Identification of quality control documents which show conformance with MIL-I-45208 or the quality control specification that is listed in the applicable instrumentation specification or acquisition document. The acceptance letter shall be referenced.

30.5 Parts list. The following information shall be presented in tabular form:

(a) Item number (corresponding to flag number identifying the part on the conformance verification drawing).
(b) Quantity of each part required assembly.
(c) Name of part with sufficient information to readily identify the part (for example: screws: thread size, length and type of head, shall be specified).
(d) Material of part.
(e) Material specification (military, federal, or commercial specification number or Government activity (NAVSEA) drawing number).

NOTE: When substitution of a material specification is made, it is the responsibility of the contractor to provide written documentation to substantiate that the substituted material is equivalent to the specified material.

(f) Type, class, grade, size, military designation, or other classification of any referenced specification.
(g) Part number or identification assigned by assembly supplier.
(h) Name of actual manufacturer of part (when applicable).
(i) Part number or identification assigned by part supplier (when applicable).
(j) Onboard repair parts. Parts that are appropriate for, or are supplied as, onboard repair parts shall be indicated by a symbol in this column.
(k) Remarks column. Finishes, platings, or coatings along with the applicable specification or other requirements should be specified in this column.

30.6 Table for special tools. Special tools required for the instrumentation shall be presented in the following tabular form:

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

(a) Item number (corresponding to flag number identifying the tool on the conformance verification drawing).
(b) Quantity of each tool required per instrument.
(c) Name (description) of tool (include generic name).
(d) Tool specification (military, federal, or commercial specification number or Government activity (NAVSEA) drawing number).
(e) Tool number or identification assigned by assembly supplier.
(f) Name of actual manufacturer of tool (when applicable).
(g) Tool number or identification assigned by tool supplier (when applicable).
(h) Description of tool's application.
(i) Remarks column. Special techniques or other usage requirements should be explained in this column.

30.7 Fluid and electrical system schematics. Fluid (piping) and electrical schematics shall be included in the verification conformance drawing when applicable. A complete schematic shall contain all the parts in the fluid or electrical systems. When confusion over its function could result, a simplified schematic containing only the major components shall also be provided. This simplified schematic may be presented in block diagram format.

(a) Complete fluid system schematic. A single schematic shall present clearly the operation and functions of the fluid system within the instrumentation. This schematic shall contain all parts (including valves, fittings, hoses, and tubing) which make up the piping or fluid system. The following features shall be incorporated into the schematic:

(1) A simplified, clear schematic in preference to one showing the physical placement or parts.
(2) A thin, broken line shall be used to represent the boundaries of each subassembly.
(3) Arrows adjacent to the parts shall indicate direction of flow.
(4) Each part shall be identified by the schematic part designation. The numbers shall be assigned in a logical sequence observing flow paths through the system.
(5) All inlet and outlet ports shall be identified.
(6) The methods of attachment for each part shall be designated by an appropriate label or symbol.
(7) The symbol used to designate a part shall be in accordance with MIL-STD-17-1 when applicable.
(8) A brief description of the fluid system operation shall be provided.
(9) Maximum operating pressure of the system and maximum pressure drop through the system (when operating at maximum pressure) shall be specified.
(10) Table for pressure rating and connection of fluid system parts. Supplementary information on parts found in the fluid (piping) system schematic shall be contained in tabular form. One column shall contain the schematic part designation for

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

each fluid system part; for example, valve V-1 and other similar designations. The other columns shall contain the following information.

(a) Item number.
(b) Description.
(c) Type (size).
(d) Pressure ratings (including operating, proof, and burst).
(e) Method of attachment, including, as applicable, brazing or welding procedure, bonding agent, and seal (gasket, O-ring, and so forth).
(f) End connections, fittings, and adapters.
(g) Relief valve ratings (include cracking and set pressure).

(b) Complete electrical system schematic. A single schematic shall represent clearly the operation and the function of the electrical circuitry within the instrumentation. The schematic shall contain all parts (including components, connectors, alarms, and so forth) which make up the electrical system. The following features shall be incorporated into the schematic:

(1) In preparation of the schematic, emphasis shall be placed on simplicity and ease of understanding of circuit operation. Physical placement of components and connecting wiring may be ignored in the interest of simplicity and clarity of this diagram.
(2) A thin, broken line shall be used to represent the boundaries of each unit or subassembly. Terminals, to which external connections are made, shall be shown within these boundaries, with the numbers, markings, type of signal, power and ground, as appropriate.
(3) Each part (such as resistors, capacitors, relays, and so forth) shall be given a unique reference designation consisting of a letter denoting the type of part (as required by ANSI/IEEE 200) and a number assigned consecutively. The numbers shall be assigned in a logical sequence of electrical current or signal flow through the circuit.
(4) In addition to where the reference designation, parts not conforming to a military specification, where permitted, shall have the following information noted adjacent to the part. Where numerical values are given, a code shall be noted to designate the units used:

(a) Resistors - Resistance, power rating, and tolerances. If variable, an arrow to indicate clockwise rotation of the control shaft.
(b) Capacitors - Capacitance, voltage rating and tolerance.
(c) Reactors - Inductance and voltage rating.
(d) Semi-conductors - type number (JEDDC number is adequate).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

(e) Integrated circuits - Operational symbol diagram of input-output relationship, terminal numbering corresponding to a representative schematic and type number.

(5) Supply voltages, phases, and frequencies and transformer terminal voltage shall be indicated and labeled as to purpose.
(6) Table for troubleshooting of electrical system. The content table shall include each test point, as identified on the electrical system schematic, with voltage, waveform or other electrical parameter that should be measured at each test point.

30.8 Selection and installation considerations.

(a) Performance data.

(1) Accuracy (in percent of span).
(2) Shock and vibration classification.
(3) Degree of water tightness of the enclosure.
(4) Electromagnetic interference and pulse susceptibility.

(b) Dimensional outline of the instrumentation showing overall and principle dimensions in sufficient detail to establish space requirements in all directions necessary for installation, servicing, exclusive of space required for operator observation of the indication.
(c) Special considerations which may affect selection or installation.

(1) Ambient temperature range.
(2) Calibration points and adjustments.
(3) Orientation.
(4) Location of instrumentation relative to vibrating equipment.
(5) Protection of the instrumentation from pulsations and spikes in the parameter being measured.
(6) Selection of the instrumentation range relative to the operating range of the system.
(7) Application for each type connection.
(8) Cleaning procedure or reference to the cleaning procedure used.
(9) Selection of the instrumentation for compatibility with the ambient environment and with the parameter being measured.

30.9 Drawing format.

(a) Unless otherwise approved by the authorized Government activity, a maximum of three sheets shall be allotted for single functioned system instrumentation and a maximum of ten sheets for instrumentation containing a multiple functioned system. A single functioned system is one that performs only one operation such as:

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

expands/contracts a pressure elastic element, steps up/steps down the voltage, conditions one electrical signal, or winds/unwinds a bimetallic element. A multiple functioned system is one that contains two or more single functioned systems.

(b) Each sheet shall be zoned.

(c) Title block shall be included on each drawing sheet and shall include the following information.

(1) Title, drawing number, and revision letter. Each sheet contain the same title, drawing number and revision letter.

(a) Title - The title shall consist of the name by which the instrumentation is known.

(b) Drawing number - The drawing number shall consist of alphanumeric characters which may be separated by dashes or slashes. The total number of characters in the drawing number (including dashes and/or slashes) shall not exceed 15. Blank spaces are not permitted within the drawing number.

(c) Revision letter - The revision letter shall denote the latest approved version of the drawing. The revision letter of conformance verification drawing shall not be changed until all the changes under that revision have been accepted in writing by the authorized Government activity (NAVSEA). No changes made to the conformance verification drawing shall be considered a revision until after the initial version of the conformance verification drawing has been accepted in writing by the authorized Government activity. After initial submittal of the conformance verification drawing, no changes shall be made during the initial drawing review process unless the change is either requested by the authorized Government activity or the change is documented in written correspondence by the contractor.

(2) Sheet _____ of ____.

(3) Tolerance on dimensions for fractions, decimals, and angles. Units of the dimensions specified on the conformance verification drawing.

(4) Contractor acceptance block (appropriate signatures and dates.)

(5) Commercial and Government Entity (CAGE).

(6) Scale.

(7) Reference drawings.

(8) Manufacturer's name and address.

(9) Drawing size.

(d) Revision block. The revision block shall be included on each sheet of the conformance verification drawing and shall contain the following information in tabular form.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.

(1) Revision letter.
(2) Description of revision.
(3) Acceptance letter serial number and originator identification.
(4) Acceptance date.

(e) Form. Sheet size and format not specified herein shall be in accordance with ANSI Y14.1.

(f) Classification designations. No Government security classification designation such as confidential or secret shall appear on the conformance verification drawing unless a particular classification is specified by the Government.

40. CONFORMANCE VERIFICATION DRAWING ACCEPTANCE

40.1 Acceptance. Acceptance shall be granted by the authorized Government activity only after the conformance verification drawing is found to meet all the requirements specified in 20. through 40.1.

50. Acquisition document instructions. The acquisition document should contain provisions that address submission, review, extension, disapproval, default, acceptance, and waiver of conformance verification drawings in addition to the effects on the delivery schedule due to delays in conformance verification drawing acceptance. NAVSEA shall be designated as the activity that accepts or disapproves the conformance verification drawing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:38Z
Check the source to verify that this is the current version before use.